



EXECUTIVE IPSEC SECURITY BRIEFING

# Executive IPsec VPN Security & Risk Posture

High-level strategic risk evaluation, cryptographic posture, and operational exposure briefing for leadership

## Section 1: Executive Summary & System Metadata

Session timestamps, endpoints evaluated, and overall audit verdict

SECURITY HARDENED PASS

NONE · COMPLIANT

Forensic inspection validates full NIST SP 800-77 compliance. AEAD encryption, authenticated DH key exchange, and sequence counter progression verified with zero defects.

### SESSION TIMESTAMPS

142.08 sec

PCAP Capture: 0.00s to 142.08s (142.08 sec)

### ORIGIN GATEWAY

Field Operations Client (Ubuntu Edge)

100.119.32.83 · StrongSwan Client Daemon

### SECURED TUNNEL

IPsec ESP over UDP 4500 (NAT-T) · AES-256-GCM / ECP-256

AES-256-GCM / ECP-256

### REMOTE GATEWAY

Enterprise Datacenter Core Gateway

100.127.207.119 · StrongSwan Core IPsec Responder

## Section 2: Holistic Risk Scorecard

Visual 0–100 score gauge, NIST compliance status badge, and deployment readiness rating

### COMPOSITE RISK INDEX



### NIST SP 800-77 REV. 1 COMPLIANCE

PASS

NIST SP 800-77 Rev. 1

### CRYPTOGRAPHIC RESILIENCE

98%

AES-GCM active, PFS missing

### COMPLIANCE ALIGNMENT

95%

NIST SP 800-77 gap identified

### DEPLOYMENT READINESS RATING

READY FOR PRODUCTION

Production readiness verified under federal cybersecurity guidelines.

### THREAT EXPOSURE INDEX

LOW

Potential retroactive decrypt

### CHANNEL STABILITY

99.9%

Zero dropped packets detected



THREAT EXPOSURE & BUSINESS IMPACT EVALUATION

# Executive Threat Matrix

Comprehensive register mapping discovered cryptographic and protocol vulnerabilities to direct business risk

Section 3: Executive Threat Matrix

Grid mapping critical vulnerabilities to business risk and operational exposure vectors

|                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Wiretap Eavesdropping &amp; Retrospective Decryption</div><div>Sev: HIGHLikelihood: Low</div><div>Adversaries recording encrypted transit can retrospectively decipher all historic corporate traffic if private keys or static PSK credentials are leaked or subpoenaed.</div><div>ACTIVE GAP</div></div>            | <div><div>Credential Cracking &amp; Identity Impersonation</div><div>Sev: MEDIUMLikelihood: Moderate</div><div>Symmetric shared keys lack individual attribution and are vulnerable to dictionary attacks, insider theft, and rogue gateway impersonation.</div><div>OBSERVED</div></div> |
| <div><div>Traffic Shape Fingerprinting &amp; Reconnaissance</div><div>Sev: MEDIUMLikelihood: High</div><div>Passive eavesdroppers observing packet length clustering and inter-arrival timing can accurately infer inner application types, operational cadences, and high-value data transfers.</div><div>OBSERVED</div></div> | <div><div>Protocol Downgrade &amp; Replay Injection</div><div>Sev: LOWLikelihood: Low</div><div>Sequence number validation prevents packet duplication and unauthorized state re-injection. Evaluated tunnel shows 100% monotonic sequence progression.</div><div>PROTECTED</div></div>   |



TRAFFIC CLASSIFICATION & STRATEGIC ROADMAP

# Traffic Overview & Strategic Remediation

Deep payload inference distribution, side-channel leakage, and prioritized resource allocation roadmap

## Section 4: High-Level Traffic Overview

Summary distribution of classified tunnel traffic and covert channel risk



## Section 5: Strategic Remediation Action Plan

Prioritized recommendations for leadership sign-off and operational resource allocation



## Executive Sign-Off & Platform Attestation



